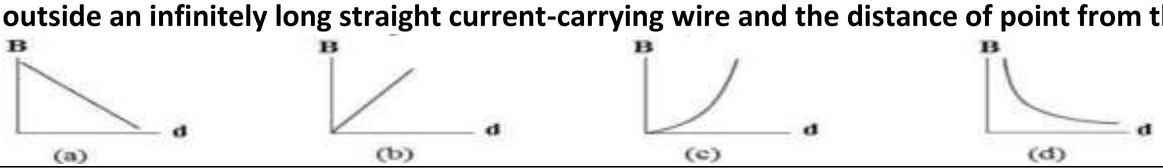
# *Unit 9*

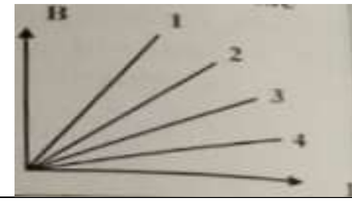
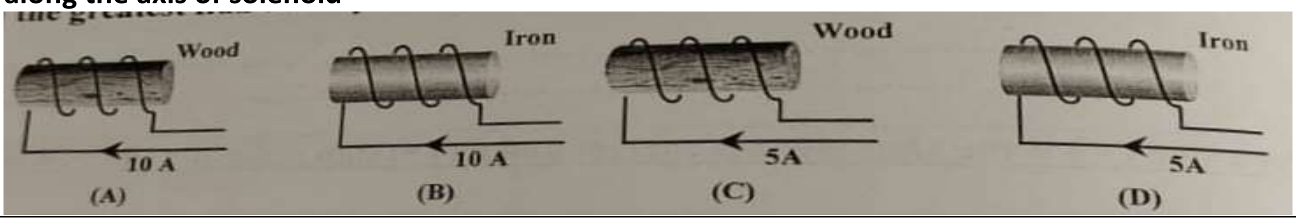
**Write the scientific term expressed by the following statements.**

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 1 | It is the total number of magnetic flux lines passing through a surface. | Magnetic flux |
| 2 | It is the point at which the total magnetic flux density vanishes. | Neutral point |
| 3 | It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field | Tesla |
| | دور اول ٢٠٢٣ | |
| 1 | It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field | Tesla |
| | دور ثاني ٢٠٢٣ | |
| 1 | The magnetic flux density which generates a force of 1 N on a straight wire of length 1m carrying current of intensity 1 A placed perpendicular in a uniform magnetic field | Tesla |
| 5 | The total number of magnetic flux lines passing through a surface and measured in weber | Magnetic flux |
| 10 | The point at which the total magnetic flux density vanishes | Neutral point |
| | تجريبي ٢٠٢٤ | |
| 1 | the total number of magnetic flux lines passing through a surface. | Magnetic flux |
| 2 | The point at which the total magnetic flux density vanishes | Neutral point |
| 3 | It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field | Tesla |
| | دور اول ٢٠٢٤ | |
| 1 | The magnetic force exerts on a wire of 1m length carrying a current of intensity 1 A and placed perpendicularly to the magnetic field | The magnetic flux density |
| 2 | The magnetic torque acting on a coil carrying current placed parallel to a uniform magnetic field of flux density of 1 Tesla | The magnetic dipole moment |
| | دور ثاني ٢٠٢٤ | |
| 1 | It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field | Tesla |
| | تجريبي ٢٠٢٥ | |
| 1 | It is a vector emanating from North pole of the coil and perpendicular to its area | The magnetic dipole moment |
| 2 | It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field | Tesla |

## Choose

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 11 | A rectangular wire loop of its dimensions (5cm * 8 cm) is at rest in a uniform magnetic field of flux density 2T that is directed into the page. The plane of the loop is perpendicular to the field, the total magnetic flux through the loop is (تجريبى)<br>a) Zero b) 0.002 Wb c) 0.0052Wb d) 0.008 Wb | D |
| 12 | If the magnitude of the magnetic field a distance d from a very long, straight, current-carrying wire is B, at what distance from the wire will the field have magnitude 4B? (تجريبى)<br>a) 8d b) 4d c) d/4 d) d/8 | C |
| 13 | The magnetic flux density increases at the center of a circular coil carrying an electric current by decreasing<br>a) cross sectional area of the coil b) the number of the coil turns<br>c) the current intensity in the coil d) the permeability of the coil core | A |
| 14 | A solenoid is carrying electric current. As an iron core is inserted into the solenoid, the magnetic field density of the electromagnet formed (تجريبى)<br>a) decreases b) increases c) remains the same d) zero | B |
| 15 | the force on a wire carrying current in a uniform magnetic field is the maximum when (تجريبى)<br>a) the current is parallel to the field lines<br>b) the current is a 60° angle W.R.T the field lines<br>c) the current at a 30° angle W.R.T the field line<br>d) the current is perpendicular to the field lines | D |
| 16 | A current carrying rectangular coil of wire is placed in a magnetic field the magnitude of the torque on the coil is not dependent upon which one of the following quantities<br>a) the magnitude of the current in the loop<br>b) the area of the loop<br>c) the direction of the current in the loop<br>d) the orientation of the loop | C |
| | دور اول ٢٠٢٣ | |
| 11 | When the density of magnetic field in a given area is increased to its double value. The magnetic flux through this area would be.........<br>(a) unchanged (b) doubled<br>(c) increased four times (d) increased eight times | B |
| 12 | Which the graph correctly gives the relation between the magnitude of the magnetic field density (B) outside an infinitely long straight current-carrying wire and the distance of point from the wire (d)?<br><br> | D |
| 13 | A solenoid is connected to a battery in a closed circuit, if its turns are compressed regularly. The magnetic flux density at a point on its axis inside it will ........<br>(a) increase (b) decrease (c) remain the same (d) vanish | A |
| 14 | A length of straight conductor, carrying a current of 3A perpendicular to a magnetic field of 0.5 T, experiences a force of 1.5 N. What is the length of the conductor?<br>(a) 0.5 m (b) 1 m (c) 2 m (d) 4 m | B |
| | دور ثاني ٢٠٢٣ | |
| 11 | The torque acting on a coil carrying current in a uniform magnetic field becomes zero when the plane of the coil is ..................... to the direction of the magnetic field.<br>a) perpendicular b) parallel<br>c) inclined by an angle 30° d) inclined by an angle 60° | A |

| | | |
|---|---|---|
| | تجريبي ٢٠٢٤ | |
| 11 | A rectangular wire loop of its dimensions (5cm * 8 cm) is at rest in a uniform magnetic field of flux density 2T that is directed into the page. The plane of the loop is perpendicular to the field, the total magnetic flux through the loop is<br>a) Zero<br>b) 0.002 Wb<br>c) 0.0052Wb<br>d) 0.008 Wb | D |
| 12 | If the magnitude of the magnetic field a distance d from a very long, straight, current-carrying wire is B, at what distance from the wire will the field have magnitude 4B? (تجريبى)<br>a) 8d b) 4d c) d/4 d) d/8 | C |
| 13 | The magnetic flux density increases at the center of a circular coil carrying an electric current by decreasing<br>a) cross sectional area of the coil b) the number of the coil turns<br>c) the current intensity in the coil d) the permeability of the coil core | A |
| 14 | A solenoid is carrying electric current. As an iron core is inserted into the solenoid, the magnetic field density of the electromagnet formed<br>a) decreases b) increases c) remains the same d) zero | B |
| 15 | the force on a wire carrying current in a uniform magnetic field is the maximum when<br>a) the current is parallel to the field lines<br>b) the current is a 60° angle with respect to the field lines<br>c) the current at a 30° angle with respect to the field lines<br>d) the current is perpendicular to the field lines | D |
| 16 | A current carrying rectangular coil of wire is placed in a magnetic field the magnitude of the torque on the coil is not dependent upon which one of the following quantities<br>a) the magnitude of the current in the loop b) the area of the loop<br>c) the direction of the current in the loop d) the orientation of the loop | C |
| | دور اول ٢٠٢٤ | |
| 5 | A square of area of cross section 10–2 m2 is placed with its axis perpendicular to a uniform magnetic field of flux density 1 T. Then the total magnetic flux through the coil is ................<br>a) $10^{-2}$ Wb b) $5*10^{-3}$ Wb c) $5\sqrt{3}*10^{-3}$ Wb d) $5*10^{-2}$ Wb | a |
| 6 | two metallic rings A and B, If a ring A of radius r , current I is flowing and in ring B of radius 2r , current 2I is flowing, then $\frac{\textit{the density of magnetic field at center of ring } A(B_A)}{\textit{the density of magnetic field at center of ring } B(B_B)}$<br>a) $\frac{1}{4}$ b) $\frac{1}{2}$ c) $\frac{1}{1}$ d) $\frac{4}{1}$ | C |
| | دور ثاني ٢٠٢٤ | |
| 8 | A solenoid carrying current has a magnetic flux density (B) at a point on its axis. If it is stretched to double its length with same current intensity, then magnetic flux density at a point on its axis is<br>A) B B) ½B C) 2B D) 4B | B |
| 9 | The magnetic flux density at a point located vertically at a distance d from a straight wire carrying a current of intensity I equals B. When increasing the current intensity to double, then the magnetic flux density becomes... A) 0.5B. B) 2B. C) B. D) 4B. | B |
| 10 | What is the magnetic field at the center of a circular loop of wire of radius 4cm when a current of 2A flows in the wire? (where : $\mu = 4\pi \times 10^{-7}$ Wb/A.m)<br>A) $\pi x\ 10^{-7}$T B) $\pi$ x $10^{-5}$ T C) $\pi x\ 10^{-4}$T D) $\pi x\ 10^{-3}$ T | B |
| | تجريبي ٢٠٢٥ | |
| 9 | A solenoid carrying current has a magnetic flux density (2B) at a point on its axis. If it is stretched to double its length with same current intensity, then magnetic flux density at a point on its axis is<br>A) B B) ½B C) 2B D) 4B | A |

| | | |
|---|---|---|
| 10 | The magnetic flux density at a point located vertically at a distance d from a straight wire carrying a current of intensity I equals X. When increasing the current intensity to double, then the magnetic flux density becomes...<br>A) 0.5X. B) 2X. C) X. D) 4X. | B |
| 21 | A circular coil of radius 0.0157 m carries a current 2 A. the magnetic flux density at the Centre of the circular coil is .... ($\mu_{air} = 4\pi*10^{-7}$ wb/Amp.m)<br>a)$1.57*10^{-3}$ T b) $8*10^{-5}$ T C) $2*10^{-3}$ T d) $3.14*10^{-1}$ T | B |
| 22 | The following graph shows the relation between the magnetic flux density B at the center of four different current-carrying circular loops .1,2,3,and 4 , and the intensity of the current passing in each of the loops, which of these four loops has the smallest radius ?<br><br>a) loop 1 b) loop 2<br>c) loop 3 d) loop 4 | A |
| 23 | Which of combination of core and current through the coil of wire will produce the greatest flux density along the axis of solenoid<br> | B |

## Mention the rule or the method used to determine

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 21 | The direction of magnetic field resulting from an electric current in a wire | Ampere right hand rule |
| 22 | The direction of magnetic field resulting from an electric current in a circular coil | The right cork screw rule (maxwell's screw rule) |
| 23 | The direction of the force (motion) of a straight wire carrying current placed in a uniform magnetic field | Fleming left hand |
| | تجريبي ٢٠٢٤ | |
| 21 | The direction of magnetic field resulting from an electric current in a wire | Ampere right hand rule |
| 22 | The direction of magnetic field resulting from an electric current in a circular coil | The right cork screw rule (maxwell's screw rule) |
| 23 | The direction of the force (motion) of a straight wire carrying current placed in a uniform magnetic field | Fleming left hand |

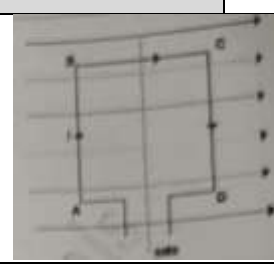
## Mention one factor that can increase

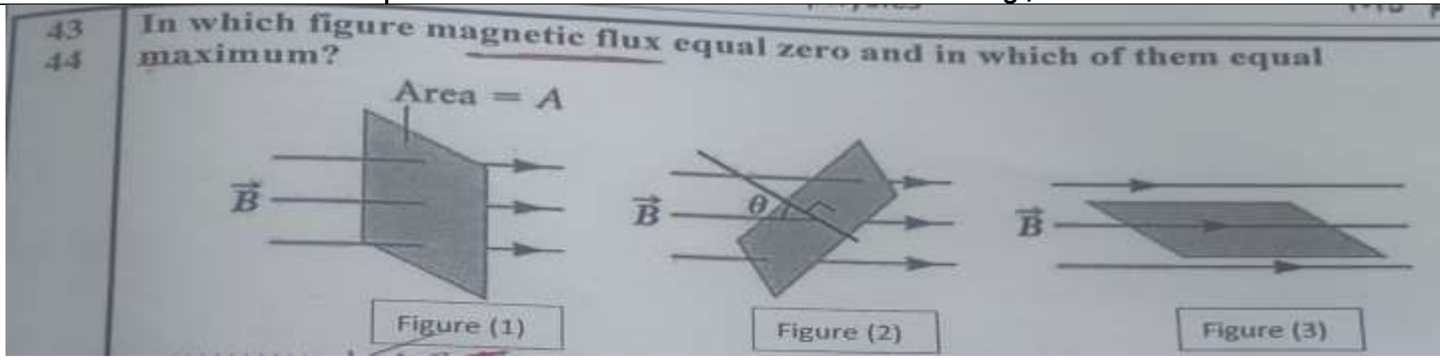
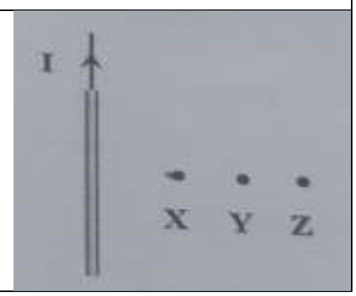
| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 27 | The density of magnetic field at a certain point from a straight wire carrying current | a) increasing intensity of current passing through the wire<br>b) decrease the distance of the point from the wire |
| 28 | The magnetic force acting on a wire carrying current placed normally in a uniform magnetic field | a) increasing the length of the wire<br>b) increase the current in the wire<br>c) increase the magnetic flux density |
| 29 | The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field | a) increasing the area of the coil's plan<br>b) increase the current intensity<br>c) increase the magnetic flux density<br>d) increase the number of turns of the coil |
| | تجريبي ٢٠٢٤ | |
| 27 | The density of magnetic field at a certain point from a straight wire carrying current | a) increasing intensity of current passing through the wire<br>b) decrease the distance of the point from the wire |
| 28 | The magnetic force acting on a wire carrying current placed normally in a uniform magnetic field | a) increasing the length of the wire<br>b) increase the current in the wire<br>c) increase the magnetic flux density |
| 29 | The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field | a) increasing the area of the coil's plan<br>b) increase the current intensity<br>c) increase the magnetic flux density<br>d) increase the number of turns of the coil |

## Mention one factor only that affects each of the following

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 39 | The density of magnetic field at a point along the axis of solenoid carrying current | a) intensity of current passing through the solenoid<br>b) number of turns c) magnetic permeability<br>d) the length of the solenoid |
| 40 | The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field | a) magnetic flux density b) the current intensity<br>c) the area of the coil's plane<br>d) number of turns of the coil |
| | دور اول ٢٠٢٤ | |
| 3 | The magnetic force acting on a wire carrying current placed perpendicular to a uniform magnetic field | a) The length of the wire (L) $F \propto L$<br>b) The current in the wire (I) $F \propto I$<br>c) The magnetic flux density (B) $F \propto B$ |
| 4 | The density of magnetic field at a point along interior axis of a solenoid carrying an electric current | a) I: intensity of current passing through the coil<br>b) L: the length of the solenoid c) μ: Permeability of medium d) N: the number of turns of wire in the solenoid |
| 47<br>48 | Mention only two factors affect The magnetic torque acting on a coil | 1) magnetic flux density ($\tau \propto B$) (at constant other factors)<br>2) the current intensity ($\tau \propto I$) (at constant other factors)<br>3) the area of the coil's plane ($\tau \propto A$) (at constant other factors)<br>4) number of turns of the coil ($\tau \propto N$) (at constant other factors)<br>5) the angle between perpendicular to coil's plane and the magnetic flux lines |

## Answer the following question

<table>
<tr><th>No.</th><th>Question</th><th>Answer</th></tr>
<tr><td></td><td colspan="2">تجريبي ٢٠٢٣</td></tr>
<tr><td>35<br>36</td><td>A circular coil of 50 turns has radius 20cm carries a current 3A. the plane of the coil makes angle $60^0$ to magnetic field whose flux density uniform of 0.5 T.<br>Calculate the magnetic torque acting the coil</td><td>$r=20*10^{-2}=0.2$ m<br>Area of circular coil= $\pi r^2 = \pi*0.2^2 = 0.125 m^2$<br>$\tau = BIAN \sin\theta =$<br>0.5*3*0.125*50*sin30= 4.71 N.m</td></tr>
<tr><td></td><td colspan="2">دور اول ٢٠٢٣</td></tr>
<tr><td>31<br>32</td><td>A rectangular coil of 500 turns has dimensions 0.2m ×0.1 m, carries a current 5A. The plane of the coil makes angle 60o to a uniform magnetic field of flux density 0.8 T .Calculate the magnetic torque acting on the coil</td><td>τ = B IAN sin θ<br>τ = 0.8 × 5 × (0.2×0.1) × 500 × sin (30)τ = 20 N.m.</td></tr>
<tr><td></td><td colspan="2">دور ثاني ٢٠٢٣</td></tr>
<tr><td>41<br>42</td><td>Calculate the intensity of electric current which passes through a circular coil of diameter 22 cm, 49 turns and produces a magnetic flux density in its center equal to $7*10^{-5}$ Tesla. (( μ= 4π x $10^{-7}$ Wb /A .m)</td><td>$B = \frac{\mu I N}{2 r}$<br>$I = \frac{B2r}{\mu N} = \frac{7*10^{-5}*2*11*10^{-2}}{4\pi*10^{-7}*49}$ =0.25 A</td></tr>
<tr><td>49<br>50</td><td>A loop of cross-sectional area 0.1 m 2 and 100 tums, carrier a current of 20 A is placed parallel to a uniform magnetic field of flux density 0.5 T. Calculate the torque acting on the loop.</td><td>τ = BIAN sin θ<br>τ = 0.5× 20× 0.1×100× sin 90 = 100 N.m</td></tr>
<tr><td></td><td colspan="2">تجريبي ٢٠٢٤</td></tr>
<tr><td>35<br>36</td><td>A circular coil of 50 turns has radius 20cm carries a current 3A. the plane of the coil makes angle $60^0$ to magnetic field whose flux density uniform of 0.5 T. Calculate the magnetic torque acting the coil</td><td>$r=20*10^{-2}=0.2$ m<br>Area of circular coil= $\pi r^2 = \pi*0.2^2 = 0.125 m^2$<br>$\tau = BIAN \sin\theta$= 0.5*3*0.125*50*sin30= 4.71 N.m</td></tr>
<tr><td></td><td colspan="2">دور اول ٢٠٢٤</td></tr>
<tr><td>7<br>8</td><td>An electric current of intensity 10 A is flowing in a long straight wire. Calculate the density of magnetic field at a distance 0.1m from the wire (Permeability of air = 4π x $10^{-7}$ Wb /A .m)</td><td>$B = \frac{\mu I}{2\pi d} = \frac{4\pi*10^{-7}*10}{2\pi*0.1} = 2*10^{-5}$ T</td></tr>
<tr><td></td><td colspan="2">دور ثاني ٢٠٢٤</td></tr>
<tr><td>45<br>46</td><td>A rectangular loop of length 0.08 m, width 0.05 m and 100 turns is placed in a magnetic field of 10 Tesla flux density, carrying current 1 A, Find The maximum torque acting on the coil</td><td>$\tau = BIAN \sin\theta =$<br>10*1*0.08*0.05*100 sin 90 = 4 N.m</td></tr>
<tr><td>N</td><td colspan="2">تجريبي ٢٠٢٥</td></tr>
<tr><td>25<br>26</td><td colspan="2">Consider a rectangular coil of wire(ABCD) carrying a current I is placed in a magnetic field of flux density (B),with its plane parallel to the field direction , as shown in the opposite figure<br><table><tr><td>Force acting on wires (AD) and (BC)</td><td>Force acting on wires (AB) and (CD)</td></tr><tr><td>Zero</td><td>Maximum</td></tr></table></td></tr>
<tr><td>27<br>28</td><td>Determine the magnetic flux density at a distance of 10cm in air from the center axis a long wire carrying a current of 10 A ($\mu_{air}$ = 4π x $10^{-7}$ Wb /A .m)</td><td>$B = \frac{\mu I}{2\pi d} = \frac{4\pi*10^{-7}*10}{2\pi*10*10^{-2}} = 2*10^{-5}$ T</td></tr>
<tr><td>39<br>40</td><td colspan="2">Mention the shape of magnetic field due to a current in a straight wire<br>a) Concentric circles whose center is the wire<br>b) The circular magnetic flux lines are closer together near the wire and farther apart from each other as the distance from the wire increases<br>c) The current passing through the wire increases, the concentric circles more crowded</td></tr>
</table>

| | |
|---|---|
| 43<br>44 | <br>Figure (1) maximum figure (3) zero |
| 45<br>46 | a rectangular loop of length 0.08 m, width 0.05 m and 100 turns is placed in a magnetic field of 10 tesla flux density , carrying current 1A , find the maximum torque acting on the coil<br>$\tau = BIAN \sin\theta = 10 * 1 * 0.08 * 0.05 * 100 * \sin 90 = 4$ N.m |
| 47<br>48 | Mention two applications on which the concept of the couple (magnetic torque) acting on the coil carrying current placed in a magnetic field is applied<br>electric motors – galvanometers |
| 49<br>50 | a long straight wire carrying current of intensity (I) as in the figure. Which one from the following points ( x,y,z) has maximum the magnetic flux density ( B)?<br>**X**<br> |

## Compare between each of the following

تجريبي ٢٠٢٣

| | | | |
|---|---|---|---|
| 49<br>50 | Point of comparison | Magnetic dipole moment | Magnetic permeability |
| | Measuring unit | $A.m^2$ or N.m/T | T.m/A OR Wb/A.m |

دور اول ٢٠٢٣

| | | | |
|---|---|---|---|
| 49<br>50 | Point of comparison | Lenz's rule | The right Cork Screw rule |
| | The use | used to determine the direction of induced current in a coil | used to determine the direction of the magnetic field resulting from an electric current in a circular coil |

دور ثاني ٢٠٢٣

| | | | |
|---|---|---|---|
| 31<br>32 | Point of comparison | Ampere's right hand rule | Fleming's left hand rule |
| | The use | It is used to determine the direction of magnetic field around a straight wire or solenoid | It is used to determine the direction of magnetic Force acting on a straight wire |
| 33<br>34 | Point of comparison | The magnetic flux density at the center of a circular loop | The magnetic flux density At point on the axis of a solenoid |
| | The physical relation used | $B = \frac{\mu I N}{2r}$ | $B = \frac{\mu I N}{L}$ |

| | دور اول ٢٠٢٤ | | |
|---|---|---|---|
| 9<br>10 | Point of comparison | Fleming's left hand rule | Ampere's right hand rule |
| | The use | It is used to determine the direction of magnetic Force acting on a straight wire carrying current | It is used to determine the direction of magnetic field around a straight wire or solenoid |

## Correct the underlined in the following statement

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 21 | The maximum value of magnetic flux cutting a certain area when the magnetic flux lines is <u>parallel</u> to that area | Perpendicular to Or Normal |
| 22 | The magnetic flux density at center of a circular coil carrying a current decrease by <u>decreasing</u> the cross-sectional area of the coil | Increasing |
| 23 | The magnetic dipole moment $\|md\|$ is given by the following relation $\|md\|$ = <u>IANsinθ</u> | I A N |
| | دور ثاني ٢٠٢٣ | |
| 21 | Existence of <u>a repulsive</u> force between parallel wires of copper each carrying current in same direction | Attractive |
| | دور ثاني ٢٠٢٤ | |
| 6 | The rule used to determine the direction of the magnetic force is called <u>right hand screw rule</u> | Fleming left hand rule |
| 7 | The force acting on a wire of length 1m carrying current of 20A placed in a magnetic field 0.1T is equal to <u>20N</u> | 2 N |
| | تجريبي ٢٠٢٥ | |
| 7 | The rule used to determine the direction of the magnetic force is called <u>right hand screw rule</u> | Fleming left hand rule |
| 8 | The force acting on a wire of length 5m carrying current of 20A placed in a magnetic field 0.01T is equal to <u>10N</u> | 1 N |

## Put sign (√ ) in front of the correct statement and sign (X) in front of the incorrect statement of each of the following:

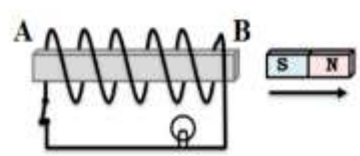
| No. | Question | Answer |
|---|---|---|
| | دور ثاني ٢٠٢٤ | |
| 2 | The force acting on a straight wire through which a current flow in a magnetic field decreases with increasing length of wire. | False |
| 3 | The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it | True |
| 4 | the magnetic torque is maximum value When the perpendicular to coil's plane normal to the magnetic field | True |
| 5 | The magnitude of the dipole moment doesn't depend on the current intensity | False |
| | تجريبي ٢٠٢٥ | |
| 3 | The force acting on a straight wire through which a current flow in a magnetic field decreases with increasing length of wire. | False |
| 4 | The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it | True |
| 5 | the magnetic torque is maximum value When the perpendicular to coil's plane normal to the magnetic field | True |
| 6 | The magnitude of the dipole moment doesn't depend on the current intensity | False |
| | | |

# Unit 10

## Write the scientific term expressed by the following statements.

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 4 | The induced current must be in a direction such as to oppose the change producing it. | Lenz's Law |
| 6 | They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. | eddy currents. |
| 9 | The ratio of output electric power to the input electric power | Efficiency of a transformer |
| 10 | It's the electromagnetic effect takes place between two coils when an induced emf generated in one of them due current variation in the other coil | (Mutual induction) |
| | دور اول ٢٠٢٣ | |
| 2 | They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. | eddy currents. |
| 3 | The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil | Self-induction |
| 4 | The ratio of output electric power to the input electric power | Efficiency of a transformer |
| | دور ثاني ٢٠٢٣ | |
| 6 | The ratio between output electric power produced from the secondary coil to the electric power given in the primary coil in the electric transformer | Efficiency of transformer |
| 8 | It is a phenomenon in which an induced electromotive force and also an induced current are generated in the conductor by changing magnetic field | Electromagnetic induction |
| | تجريبي ٢٠٢٤ | |
| 4 | The induced current must be in a direction such as to oppose the change producing it. | Lenz's Law |
| 5 | They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. | eddy currents. |
| 9 | The ratio of output electric power to the input electric power | Efficiency of a transformer |
| 10 | It's the electromagnetic effect takes place between two coils when an induced emf generated in one of them due current variation in the other coil | (Mutual induction) |
| | دور اول ٢٠٢٤ | |
| 11 | In the transformer The ratio of output electric power to the input electric power | Efficiency of a transformer |
| 12 | A phenomenon in which an induced electromotive force and also an induced current are generated in the conductor by changing magnetic field( magnetic flux) + | Electromagnetic induction |
| | دور ثاني ٢٠٢٤ | |
| 17 | induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. | eddy currents. |
| 18 | The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil | Self-induction |
| | تجريبي ٢٠٢٥ | |
| 17 | induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. | eddy currents. |
| 18 | The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil | Self-induction |

## Choose

| No. | Question | Ans |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 17 | which of the following will cause an induced current in a coil of wire<br>a) A magnet resting near the coil.    b) the constant field of the earth passing through the coil<br>c) A magnet being moved into or out of the coil.    d) A wire carrying a constant current near the coil. | C |
| 18 | A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is perpendicular to the magnetic field is ( V ). If the frequency of rotation is halved, then the instantaneous value of emf produced at the same position is………..    (a) 1/4 V    (b) 1/2 V    (c) V    (d) 2 V | B |
| 20 | If a transformer is assumed to be an ideal one , all the following statements is correct EXCEPT<br>(a) The current ratio of secondary to primary is equal to The turn's ratio of secondary to primary<br>(b) The current ratio of secondary to primary is equal to The turn's ratio of primary to secondary<br>(c) The voltage ratio of secondary to primary is always equal to The turn's ratio of secondary to primary<br>(d) The electric power produced at the secondary coil is equal to electric power given in the primary coil | A |
| | دور اول ٢٠٢٣ | |
| 15 | A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is parallel to the magnetic field is (100 V). If the frequency of rotation is doubled, then the instantaneous value of emf produced at the same position is………..    (a) 50V    (b) 100V    (c) 200V    (d) 400V | C |
| 20 | In an ideal step up transformer, which of the following statements is incorrect<br>(a) The current ratio of secondary to primary is smaller than one.<br>(b) The turn's ratio of primary to secondary is smaller than one.<br>(c) The voltage ratio of primary to secondary is greater than one.<br>(d) The output-to-input electric power ratio is smaller than one | D |
| | دور ثاني ٢٠٢٣ | |
| 12 | When a coil rotates in a magnetic field, the direction of the induced electromotive force produced in the coil is changed each ……………cycle.<br>A) 1/4    B) 1/2    C) 1    D) 3/4 | B |
| 13 | When the number of turns in the dynamo's coil is doubled at constant other factors then the maximum electromotive force generated from it……………<br>a) Increases to the double    b) decreases to half of its value<br>c) remains constant    d) Increases three times of its value | A |
| 14 | In the opposite figure, when the magnet moves in the direction shown the pole formed at A is …<br>a) North<br>b) South<br>c) west<br>d) east<br> | B |
| | تجريبي ٢٠٢٤ | |
| 17 | which of the following will cause an induced current in a coil of wire<br>a) A magnet resting near the coil.    b) the constant field of the earth passing through the coil<br>c) A magnet being moved into or out of the coil.    d) A wire carrying a constant current near the coil. | C |
| 18 | A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is perpendicular to the magnetic field is ( V ). If the frequency of rotation is halved, then the instantaneous value of emf produced at the same position is………..    (a) 1/4 V    (b) 1/2 V    (c) V    (d) 2 V | B |

**20** If a transformer is assumed to be an ideal one , all the following statements is correct EXCEPT
(a) The current ratio of secondary to primary is equal to The turn's ratio of secondary to primary
(b) The current ratio of secondary to primary is equal to The turn's ratio of primary to secondary
(c) The voltage ratio of secondary to primary is always equal to The turn's ratio of secondary to primary
(d) The electric power produced at the secondary coil is equal to electric power given in the primary coil

Answer: A

## دور اول ٢٠٢٤

**15** A straight wire 1m long is moving in a uniform magnetic field of flux density 0.1T at a uniform velocity 10m/s. an induced potential difference of 0.5V is generated between its terminals. The angle between the direction of the wire motion and the direction of the magnetic field is.
a) 0° b) 30° c) 60° d) 90°

Answer: b

**16** A coil which carries current I is wound on iron core. The self-induced electromotive force in the coil is not affected by………
a) Variation in coil current b) Variation in voltage to the coil
c) Change of number of turns of coil d) The resistance of coil wire

Answer: d

**17** According to the Lenz's law,……………….
a) The direction of the induced emf will be such that it helps the change in magnetic flux
b) The direction of the induced current will be such that it opposes the change in magnetic flux.
c) The magnitude of the induced electromotive force (emf) is directly proportional to the rate of change of flux.
d) The direction of the induced electromotive force (emf ) will be along the direction of the magnetic field.

Answer: B

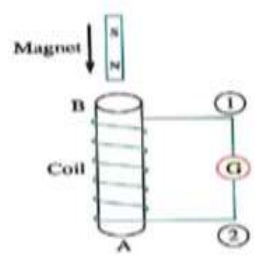
## دور ثاني ٢٠٢٤

**11** When the number of turns in the dynamo's coil is doubled, then the maximum generated electromotive force generated from it:
a) Increases to double b) Decreases to its half value
c) Remained constant d) Increases 4 times

Answer: A

**12** A bar magnet is dropped into a solenoid connected to a galvanometer, as shown in the opposite figure. Which choice of the following is correct?

| | The magnetic pole formed at A | The magnetic pole formed at B |
|---|---|---|
| a) | North | North |
| b) | South | South |
| c) | South | North |
| d) | North | South |




Answer: C

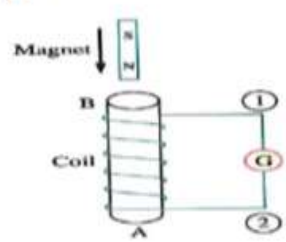
## تجريبي ٢٠٢٥

**11** When the number of turns in the dynamo's coil is doubled, then the maximum generated electromotive force generated from it:
a) Increases to double b) Decreases to its half value
c) Remained constant d) Increases 4 times

Answer: A

**12** A bar magnet is dropped into a solenoid connected to a galvanometer, as shown in the opposite figure. Which choice of the following is correct?

| | The magnetic pole formed at A | The magnetic pole formed at B |
|---|---|---|
| a) | North | North |
| b) | South | South |
| c) | South | North |
| d) | North | South |




Answer: C

| | | |
|---|---|---|
| 24 | In Faraday's Experiment for electro-magnetic induction, the induced emf in the coil increase when:<br>a) keeping the magnet still inside the coil<br>b) speeding up the magnet motion relative to the coil<br>c) connecting a galvanometer to the coil<br>d) increasing the spacing between the coil turns | B |
| 35<br>36 | A rectangular coil of 100 turns and cross-sectional area 0.06 $m^2$ rotates at frequency 50Hz in a uniform magnetic field of flux density 0.1 T. The maximum induced emf is…<br>a) 30v b) 94.2 v c) 188.5 v d) 60 v | C |

## Mention the rule or the method used to determine

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 24 | The direction of induced current in moving wire in a uniform magnetic field | Fleming right hand rule |
| 26 | Production of backward induced electromotive force (emf) on secondary coil when the circuit of primary coil is closed | A) increasing the current intensity in primary coil by using rheostat<br>b) moving the primary coil near to or inside secondary coil |
| | تجريبي ٢٠٢٤ | |
| 24 | The direction of induced current in moving wire in a uniform magnetic field | Fleming right hand rule |
| 26 | Production of backward induced electromotive force (emf) on secondary coil when the circuit of primary coil is closed | A) increasing the current intensity in primary coil by using rheostat<br>b) moving the primary coil near to or inside secondary coil |

## Mention one factor that can increase

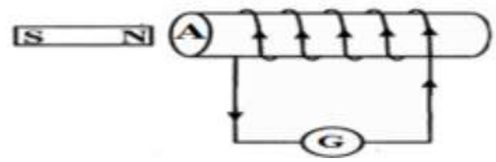
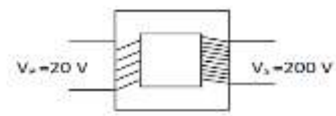
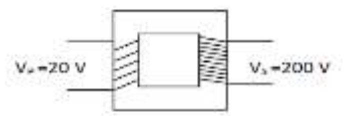
| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 30 | The induced emf in a straight wire moving normal to a magnetic field | a) increase the length of wire<br>b) increase the density of the magnetic field<br>c) increase velocity of motion |
| 31 | The induced voltage produced from the electric transformer | a) by using step up transformer<br>b) by making the transformer ratio more than one |
| | تجريبي ٢٠٢٤ | |
| 30 | The induced emf in a straight wire moving normal to a magnetic field | a) increase the length of wire<br>b) increase the density of the magnetic field<br>c) increase velocity of motion |
| 31 | The induced voltage produced from the electric transformer | a) by using step up transformer<br>b) by making the transformer ratio more than one |

## Mention one factor only that affects each of the following

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 41 | The mutual induction coefficient between two coils | a) The presence of an iron core inside the coil.<br>b) The volume of the coil and the number of its turns.<br>c) The distance separating them |
| 42 | The induced emf in a straight wire moving normal to a magnetic field | a) the length of wire ($l$)<br>b) density of the magnetic field (B)<br>c) velocity by which the wire move (v) |

**Answer the following question**

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 33<br>34 | **in an ideal transformer, the secondary coil has twice as many turns as the primary. If the alternating current in the primary coil is 2A. calculate the current in the secondary coil.**<br>$N_p / N_s = I_s / I_p$ $N_p / 2N_P = I_s / 2$ $I_s = 1A$ | |
| 37<br>38 | **A square loop of 10 turns is placed in a uniform magnetic field perpendicular to the plane of the loop, if the magnetic flux changes from 0.03 Wb to zero in 0.1 s. Calculate the average induced emf in the loop**<br>emf = -N * (ΔΦ / Δt) $\Delta\Phi = \Phi_2 - \Phi_1 = 0 - 0.03 = -0.03$ Wb emf = -10 * (-0.03 / 0.1) = 3 V | |
| 39<br>40 | **when the current changes from +2A to -2A in 0.05 sec, an emf induced in a coil 8V ,Calculate the coefficient of self-induction in the coil.**<br>$emf = -L\frac{\Delta I}{\Delta t}$ $L = -\frac{emf\Delta t}{\Delta I} = \frac{-8 \times 0.05}{-4} = 0.1\ \mathrm{H}$ | |
| | دور اول ٢٠٢٣ | |
| 29<br>30 | **When the current in a coil changes from 5A to 2A in 0.05 s, an e m f of 15 V is induced in the coil. Calculate the coefficient of self-induction of the coil**<br>emf = –L × Δ I/Δ t 15 = –L × (2 – 5)/ 0.05 L = 0.25 H | |
| 33<br>34 | **A metallic ring is placed in a uniform magnetic field perpendicular to the plane of the ring. If the magnetic flux changes from 0.05 Wb to – 0.05 in 0.05 s. Calculate the average induced EMF in the loop. (دور أول 2023)**<br>emf = -N * (ΔΦ / Δt) we're dealing with a single ring, the number of turns, N, is 1.<br>$\Delta\Phi = \Phi_2 - \Phi_1 = -0.05 - 0.05 = -0.1$ Wb emf = -1 * (-0.1 / 0.05) = 2 V | |
| | دور ثاني ٢٠٢٣ | |
| 43<br>44 | **An ideal step-down transformer is used to operate an electric lamp, which works on a potential difference of 30 volts. If an electric source of 240 volts is used & the number of tums of the primary coil is 480 turns, calculate the number of tums of the secondary coils.**<br>Np / Ns = Vp / Vs 480 / Ns = 240 V / 30 V 480 / Ns = 8 Ns = 480 / 8 = 60 turns | |
| 45<br>46 | **An electric current of intensity 5 A flowing through a coil of self-inductance 0.001 H. If the current vanishes in 0.5 second, calculate The induced electromotive force (emf) generated in the coil.**<br>$emf = -\frac{L\Delta I}{\Delta t} = \frac{-0.001 \times (0-5)}{0.5} = 0.01\ \mathrm{V}$ | |
| | تجريبي ٢٠٢٤ | |
| 33<br>34 | **in an ideal transformer, the secondary coil has twice as many turns as the primary. If the alternating current in the primary coil is 2A. calculate the current in the secondary coil.**<br>$N_p / N_s = I_s / I_p$ $N_p / 2N_P = I_s / 2$ $I_s = 1A$ | |
| 37<br>38 | **A square loop of 10 turns is placed in a uniform magnetic field perpendicular to the plane of the loop, if the magnetic flux changes from 0.03 Wb to zero in 0.1 s. Calculate the average induced emf in the loop**<br>emf = -N * (ΔΦ / Δt) $\Delta\Phi = \Phi_2 - \Phi_1 = 0 - 0.03 = -0.03$ Wb emf = -10 * (-0.03 / 0.1) = 3 V | |
| 39<br>40 | **when the current changes from +2A to -2A in 0.05 sec, an emf induced in a coil 8V ,Calculate the coefficient of self-induction in the coil.**<br>$emf = -L\frac{\Delta I}{\Delta t}$ $L = -\frac{emf\Delta t}{\Delta I} = \frac{-8 \times 0.05}{-4} = 0.1\ \mathrm{H}$ | |
| | دور اول ٢٠٢٤ | |
| 43<br>44 | **A metallic ring is placed in a magnetic field which changes from 0.004 Wb to 0.008 Wb in 2s. Calculate the induced electromotive force( emf) in the coil**<br>emf = -N * (ΔΦ / Δt) = $\frac{-1*(0.004-0.008)}{2}$ = $2*10^{-3}$ v | |

| | |
|---|---|
| 49<br>50 | In the opposite diagram a magnet is moving relative to the solenoid , look well at the opposite diagram then determine the following<br>a) kind of the magnetic pole formed at A<br>b) The direction of the magnet motion<br><br>a ) north pole (N)<br>b) the magnet is moving towards the coil |
| | دور ثاني ٢٠٢٤ |
| 49<br>50 | Is this step-up or step-down transformer?<br>step up because $V_s > V_p$<br>or $N_S > N_p$<br> |
| | تجريبي ٢٠٢٥ |
| 41<br>42 | Is this step-up or step-down transformer? Why?<br>step up<br>because $V_s > V_p$ or $N_S > N_p$<br> |

## Compare between each of the following

تجريبي ٢٠٢٣

| | Point of comparison | Electric generator | Electric motor |
|---|---|---|---|
| 43<br>44 | The scientific base | Electromagnetic induction | Magnetic torque acting on a coil carrying current placed in a magnetic field |

| | Point of comparison | Direct current (DC) | Alternating current (AC) |
|---|---|---|---|
| 47<br>48 | The direction | One direction | Two opposite directions |

| | Point of comparison | Cathode ray tube | The electric transformer |
|---|---|---|---|
| 45<br>46 | The use | Used in TV and computer monitors | Converting low alternating voltage into high alternating voltage and vice versa Or it transfers electric power from one circuit to another |

دور اول ٢٠٢٣

| | Point of comparison | Direct current (DC) | Alternating current (AC) |
|---|---|---|---|
| 47<br>48 | The direction | One direction | Two opposite directions |

| | Point of comparison | Lenz's rule | The right Cork Screw rule |
|---|---|---|---|
| 49<br>50 | The use | used to determine the direction of induced current in a coil | used to determine the direction of the magnetic field resulting from an electric current in a circular coil |

دور ثاني ٢٠٢٤

| | Point of comparison | Fleming's right-hand rule | Lenz's rule |
|---|---|---|---|
| 15<br>16 | The use | It is used to determine the direction of induced current in the wire | used to determine the direction of induced current in a coil |

تجريبي ٢٠٢٥

| | Point of comparison | Fleming's right-hand rule | Lenz's rule |
|---|---|---|---|
| 15<br>16 | The use | It is used to determine the direction of induced current in the wire | used to determine the direction of induced current in a coil |

| | Point of comparison | ac generator | Transformer |
|---|---|---|---|
| 37<br>38 | Working principle | Electromagnetic induction | Mutual induction |

## Correct the underlined in the following statement

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 24 | The mutual induction coefficient measured in Ampere/second | H. Wb/Amp or Ohm. Sec or Volt.sec/Amp |
| | دور ثاني ٢٠٢٣ | |
| 22 | The electromotive force induced in a rectangular coil rotating in a magnetic field becomes zero when the coils' plane is parallel to magnetic field | Perpendicular or normal |
| 24 | Maximum induced emf in a straight wire moving parallel to a magnetic field | Perpendicular or normal |
| 28 | The coefficient of mutual inductance between two coils increases as the rate of change of current in the primary coil decreases | Constant / doesn't change |
| | دور اول ٢٠٢٤ | |
| 18 | A laminated iron core is used in an electric transformer to increase eddy currents | Decrease or minimize) |
| 19 | The ac generator is a device based on the principle of magnetic torque | Electromagnetic induction |
| 20 | The induced electromotive force (emf) generated in a coil increases to double if the rate of change of current in the same coil increases 4 times | 4 times |
| | تجريبي ٢٠٢٥ | |
| 31 | There is an induced emf in the secondary coil only when the current in the primary coil is constant | Change/changeable /variable |
| 32 | Switching on the current in primary coil or increasing the current intensity in primary coil by using rheostat, producing forward induced emf | Back induced emf |
| 33 | Alternating current (AC) has constant direction and its intensity periodically changed | Two direction / opposite |
| 34 | The ac generator is a device used for converting heat energy into electrical energy | Mechanical |

## Mention the mathematical formula for each of the following

| No. | Question | Answer |
|---|---|---|
| | دور ثاني ٢٠٢٤ | |
| 13 | Mutual induction coefficient | $M = emf. \Delta t / \Delta I$ Or $emf = M \Delta I / \Delta t$ |
| 14 | average induced emf in a coil | $Emf = - N \Delta \phi / \Delta t$ |
| | تجريبي ٢٠٢٥ | |
| 13 | Mutual induction coefficient | $M = emf. \Delta t / \Delta I$ Or $emf = M \Delta I / \Delta t$ |
| 14 | average induced emf in a coil | $Emf = - N \Delta \phi / \Delta t$ |

## A) Put sign (√ ) in front of the correct statement and sign (X) in front of the incorrect statement

| No. | Question | Ans |
|---|---|---|
| | دور ثاني ٢٠٢٤ | |
| 19 | The ratio of output electric power to the input electric power in the electric transformer is called efficiency of transformer | True |
| 20 | If the wires moves by an angle along the direction of the magnetic field an induced emf generated between two ends of the wire is given by : $emf = BLV / \sin \theta$ | False |
| | تجريبي ٢٠٢٥ | |
| 19 | The ratio of output electric power to the input electric power in the electric transformer is called efficiency of transformer | True |
| 20 | If the wires moves by an angle along the direction of the magnetic field an induced emf generated between two ends of the wire is given by : $emf = BLV / \sin \theta$ | False |
| 29 | From lenz's rule, the direction of induced current in generated in the conductor depends on the direction of the motion and the direction of the magnetic field | True |
| 30 | When the magnet is moved away from the solenoid, magnetic flux link with the coil increases, the coil near end forms like pole which tends to repel it | False |

# Unit 11

**Write the scientific term expressed by the following statements.**

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 6 | The distribution of the radiation intensity with wavelength | Planck's distribution |
| 7 | The wavelength at maximum radiation intensity (λ max) is inversely proportional to the temperature | Wien's Law |
| 8 | emission of electrons from a heated metal's surface (by heating) | Thermionic emission |
| | دور اول ٢٠٢٣ | |
| 5 | It is an ideal system that absorbs all radiation falling on it. Then re-emit this radiation totally again | Black body radiation |
| 6 | It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it | Surface potential barrier |
| 7 | It is the minimum energy needed to free (release) an electron from the surface of the metal | Work function |
| 8 | They are invisible electromagnetic waves of short wavelengths, located between gamma rays and UV rays | X- ray |
| 9 | An external photon of energy E2 – E1 passes by excited atom at E 2 to relax it to E1 before life time is over | Stimulated emission |
| | دور ثاني ٢٠٢٣ | |
| 2 | Light Amplification by Stimulated Emission of Radiation | LASER |
| 3 | The law which states that the wavelength (λm) at maximum radiation intensity is inversely proportional to temperature on kelvin scale | Wien's law |
| 7 | Emission of electrons from a metallic surface when the light energy falling on it | Photoelectric effect |
| | تجريبي ٢٠٢٤ | |
| 6 | The distribution of the radiation intensity with wavelength | Planck's distribution |
| 7 | The wavelength at maximum radiation intensity (λ max) is inversely proportional to the temperature | Wien's Law |
| 8 | emission of electrons from a heated metal's surface (by heating) | Thermionic emission |

## Choose

| No. | Question | Ans |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 19-For a black body radiation if $\lambda_{max}$ represents the maximum wavelength corresponding to maximum radiation at temperature T, the curve between $\lambda_{max}$ and T is …………… $\lambda_{max}$  a. A b. B c. C d. D | | d |
| | دور اول ٢٠٢٣ | |
| 16- A black body is heated to a temperature of 3000 K so the wavelength that is associating with its radiation of the maximum intensity was (λ). If it is cooled to an absolute temperature (T) then the wavelength associating with its radiation of the maximum intensity became (10 λ) so the temperature (T) is ……… (a) 300 K (b) 2700 K (c) 270 K (d) 1800 K | | A |
| 18- All of the following are from the properties of photons Except……… (a) they travel in straight line with constant speed equals the speed of light. (b) they have linear momentum(PL). (c) they can be deflected under the effect of electric and magnetic fields. (d) they travel in space in the form discrete packets of energy (quanta) | | C |

**19- A monochromatic light falls on a metallic surface, the electrons with kinetic energy are emitted when**
**(a) the frequency of incident photon decreases**
**(b) the energy of the incident photon is greater than the work function of the metal**
**(c) the energy of the incident photon is less than the work function of the metal**
**(d) the energy of the incident photon equal to the work function of the metal**

**B**

## دور ثاني ٢٠٢٣

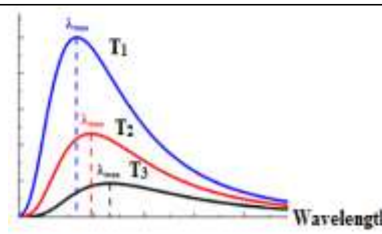
**16- In Compton effect, when a photon of X-rays collides with a free electron initially at rest, the photon energy after collision …….**
**a) Increases b) decreases c) remain the same d) vanishes**

**B**

**17- When emitted photons propagate having the same phase, it is said that they are**
**a) Spreading b) collimated c) monochromatic d) coherent**

**D**

**18-The plots of intensity of radiation versus wave length of three black bodies at temperatures T1, T2 and T3 are shown. Then**
**a)T3 > T2 > T1**
**b) T2 > T3 > T1**
**c) T1 > T3 > T2**
**d) T1 > T2 > T3**




**D**

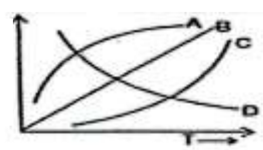
**19- Using x-rays in studying the crystalline structure of materials because it …**
**a) Faster than light b) Invisible**
**c) Diffract through atoms d)ionizing radiation**

**C**

## تجريبي ٢٠٢٤

**19- For a black body radiation if $\lambda_{max}$ represents the maximum wavelength corresponding to maximum radiation at temperature T, the curve between $\lambda_{max}$ and T is ……………**

λ




**a. A b. B c. C d. D**

d

## دور اول ٢٠٢٤

**24)The process in which electrons are emitted from a metallic surface when heat given to it is called………………**
**a) Photoelectric effect b) Thermionic emission**
**c) Compton effect d) Black body radiation**

**B**

**25)The common feature of laser and (x) rays photons are that they……..**
**a) are coherent b) are monochromatic**
**c) have the same speed d) have the same energy**

**c**

**26)To increase the number of photoelectrons emitted per second when light falling on a metallic surface, …….**
**a) The frequency of incident photon increases**
**b) Kinetic energy of emitted electron increases**
**c) The wavelength of incident photon increases**
**d) The intensity of the radiation increases**

**d**

## دور ثاني ٢٠٢٤

**26-The energy of the cathode rays equals …………**
**a. h υ b. $1/2\ mV^2$ c. mV d. 2 mV**

**B**

**27-The ratio between the frequencies of photons is 1 : 2. So, the ratio between their energies is a. 1 : 1 b. 2 : 1 c. 1 : 2 d. 4 : 1**

**c**

**28-When a photon collides with a free electron so, ………….**
**a. the photon loses its energy.**
**b. the electron acquires kinetic energy equals the photon energy.**
**c. the frequency of the scattered photon is less than the frequency of the incident photon.**

**C**

| d. the photon and the electron move together. | |
|---|---|

## Mention the rule or the method used to determine

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 25 | Image moving objects in the dark due to the heat radiation which these objects re-emit | Thermal imaging (remote sensing) night vision system |
| | تجريبي ٢٠٢٤ | |
| 25 | Image moving objects in the dark due to the heat radiation which these objects re-emit | Thermal imaging (remote sensing) night vision system |

## Mention one factor that can increase

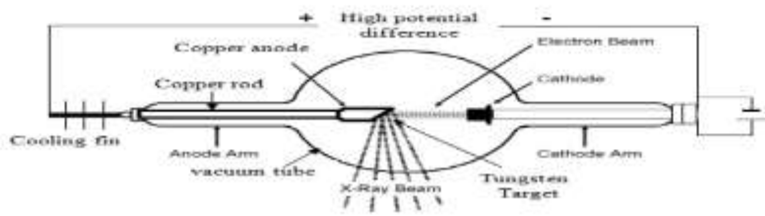
| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 32 | The velocity of electron emitted from cathode ray tube | Increasing the potential difference between the cathode and anode |
| | تجريبي ٢٠٢٤ | |
| 32 | The velocity of electron emitted from cathode ray tube | Increasing the potential difference between the cathode and anode |

## Mention one factor only that affects each of the following

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 43 | The velocity of electron emitted from the cathode in cathode ray tube | the potential difference between the cathode and anode |
| 44 | The minimum wavelength in the continuous radiation of X-rays | the potential difference between the cathode and anode |
| | دور اول ٢٠٢٤ | |
| 27 | The wavelength of the characteristic X-rays. | the kind of target element<br>or the atomic number of the target element |
| 28 | The wave length at maximum radiation intensity in Planck's curve | The temperature of the hot body |

## Answer the following question

| No. | Question | Answer |
|---|---|---|
| | تجريبي ٢٠٢٣ | |
| 41<br>42 | A hot body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun<br>$\frac{\lambda 1}{\lambda 2} = \frac{T2}{T1}$ $= \frac{20000}{5000} = \frac{T2}{1500}$ $T_2 = \frac{20000 \times 1500}{5000}$ $T_2 = 6000K$ | |
| | دور اول ٢٠٢٣ | |
| 35<br>36 | A metal has a work function $3.85 \times 10^{-19}$ J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency $7.5 \times 10^{14}$ Hz is shone on the surface (Given that $h = 6.6 \times 10^{-34}$ J.s)<br>K. E = hν – Ew   $K.E = (6.6 \times 10^{-34} \times 7.5 \times 10^{14}) - (3.85 \times 10^{-19})$ $K.E = 1.1 \times 10^{-19}$ J | |
| | تجريبي ٢٠٢٤ | |

| | | |
|---|---|---|
| 41<br>42 | A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun<br>$\frac{\lambda 1}{\lambda 2} = \frac{T2}{T1} = \frac{20000}{5000} = \frac{T2}{1500}$ T2= $\frac{20000\times1500}{5000}$ = 6000K | |
| | دور اول ٢٠٢٤ | |
| 41<br>42 | A photon of wavelength $5.6\times10^{-7}$ m falls on a metallic surface of work function $3.42\times10^{-19}$ J, calculate the kinetic energy of emitted electron from the metallic surface .(Planck's constant = $6.625\times10^{-34}$ J.s, speed of light in air $=3\times10^{8}$ m/s) | $K.E_{max} = h\nu - E_w$<br>$K.E_{max} = \frac{hc}{\lambda} - E_w = \frac{6.625\times10^{-34}\times3\times10^{8}}{5.6\times10^{-7}} - 3.42\times10^{-19}$<br>$K.E_{max} = 1.29\times10^{-20}$ J |
| 45<br>46 | Mention without explanation the two conditions necessary for laser action | 1) Population inversion<br>2) Stimulated emission |
| | دور ثاني ٢٠٢٤ | |
| 24<br>25 | In the opposite figure: Mention the name of the device and mention one use<br>Coolidge tube<br>Used to produce X-rays |  |
| 41<br>42 | Mention two properties of LASER | Coherence – collimated – monochromaticity -intensity |
| 43<br>44 | If the critical frequency of zinc is $3*10^{8}/3000*10^{-10}$ Hz. Calculate its work function.(Planck's constant = $6.625\times10^{-34}$ J.s) | $E_W = h\,\nu = 6.625\times10^{-34} * 3*10^{8}/3000*10^{-10} = 6.625*10^{-19}$ J |

## Compare between each of the following

| | | | |
|---|---|---|---|
| | تجريبي ٢٠٢٣ | | |
| 45<br>46 | Point of comparison | Cathode ray tube | The electric transformer |
| | The use | Used in TV and computer monitors | Converting low alternating voltage into high alternating voltage and vice versa<br>Or it transfers electric power from one circuit to another |
| | دور ثاني ٢٠٢٣ | | |
| 35<br>36 | Point of comparison | Ordinary light | LASER |
| | monochromaticity | Non monochromatic | monochromatic |
| 39<br>40 | Point of comparison | The continuous spectrum | The line spectrum |
| | dependence on voltage between filament and target material | Depends | Not depend |

## Correct the underlined in the following statement

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 25 | The wavelength of the scattered photon (λ scattered) is <u>smaller than</u> the wavelength of the incident photon (λ incident) | Greater than |
| 26 | The wavelength of the characteristic radiation of X- rays <u>increases</u> as the atomic number of target material increases | Decreases |

| | | |
|---|---|---|
| 27 | The ordinary light is <u>not subject</u> to the inverse square law | Subject (obey) |
| | دور ثاني ٢٠٢٣ | |
| 23 | Stimulated emission is dominant in <u>ordinary light</u> sources | LASER |
| 25 | Emission of photoelectron with kinetic energy from metal occurs when the energy of incident radiation <u>smaller</u> than the work function of the metal | Greater |
| 26 | The minimum wavelength of the continuous radiation decreases when the potential difference between the cathode and the anode <u>decreases</u> | Increases |
| 30 | Function of <u>Grid</u> in cathode ray tube (CRT) is heating up the cathode | Filament |
| | دور ثاني ٢٠٢٤ | |
| 21 | The wavelength which is associating the maximum intensity of the radiation (λm) is <u>directly</u> proportional to the temperature of a radioactive source on Kelvin scale. | inversely |
| 22 | Wavelengths of invisible electromagnetic rays lie between the wavelengths of ultraviolet rays and gamma rays and is characterized by a short wavelength is called <u>infrared</u> rays | X – ray |
| 23 | metastable energy level characterized by its lifetime is relatively long nearly <u>$10^{-8}$ s</u> | $10^{-3}$ s |

A) Put sign (√ ) in front of the correct statement and sign (X) in front of the incorrect statement

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٤ | |
| 21 | At low temperatures, the wavelengths of the thermal radiation are mainly in the infrared region | True |
| 22 | In Compton Effect the frequency of the scattered photon is greater than the frequency of the incident photon | False |
| 23 | The maximum frequency of the continuous radiation of X-rays does not depend on the potential difference between the cathode and anode | False |
| | دور ثاني ٢٠٢٤ | |
| 29 | To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be less than the work function | False |
| 30 | The Scientific idea of cathode ray tube is depended on the thermionic emission | True |

## Write the mathematical law used to calculate

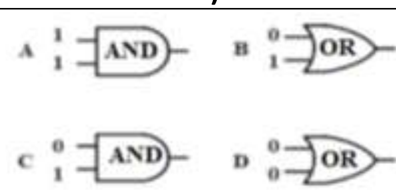
| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٤ | |
| 29 | Momentum of photon | $P_L = E/C = h/\lambda -= h\upsilon / C = m C$ |
| 30 | Kinetic energy of emitted electron when light falling on a metallic surface. | $K.E_{max} = h\nu - E_w = 1/2\ m V^2$ |

# *Unit 12*

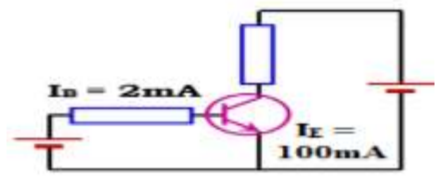
**Write the scientific term expressed by the following statements.**

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 10 | The circuits that perform logic operations and depend on digital electronics | Logic gates |
| | دور ثاني ٢٠٢٣ | |
| 4 | The number of bonds broken per second will be equal to the number of bonds mended per second | Thermal equilibrium or dynamic equilibrium |
| 9 | Digital circuits that perform logic operations. | Logic gates |
| | دور اول ٢٠٢٤ | |
| 31 | The number of bonds broken per second will be equal to the number of bonds mended per second | Thermal equilibrium or dynamic equilibrium |
| 32 | The ratio between the collector current and the base current | Current gain |

## Choose

| No. | Question | Ans |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 17 | Which of the following statements is correct when connecting the diode in reverse direction?<br>(a) The width of depletion layer increases and diode resistance decreases.<br>(b) The width of depletion layer decreases and diode resistance decreases.<br>(c) Neither the width of depletion layer nor the diode resistance changes.<br>(d) The width of depletion layer increases and the diode resistance increases | d |
| | دور ثاني ٢٠٢٣ | |
| 15 | When the temperature of pure silicon crystal (Si) rises, so its electrical conductivity.......<br>a) decreases b) does not change c) increases d) vanishes | C |
| 20 | Which of the following gates its output is 1?<br>a) Only B b) Only D<br>c) Only A d) A,B<br> | D |
| | دور اول ٢٠٢٤ | |
| 33 | Doping a semiconductor with impurities of ............ atoms creates an n-type material.<br>a) trivalent b) tetravalent c) pentavalent d) divalent | C |
| 34 | When connecting the p-n Junction (diode) in reverse biasing, then the width of depletion layer........<br>a) increases b) decreases c) does not change d) becomes zero | A |
| 35 | Which of the following gates that if both inputs are low the output is high?<br>a) OR gate b) AND gate c) NAND gate d) NOT gate | C |

## Answer the following question

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 37<br>38 | The current gain (β) of a transistor in common emitter mode is 50. Calculate the collector current if the base current 50µ A<br>$\beta = IC / IB$ $50 = IC / 50$ $IC = 2500\ \mu A = 2.5 \times 10^{-3}\ A$ | |
| | دور ثاني ٢٠٢٣ | |
| 47<br>48 | **npn transistor is used as a current amplifier, if the base current equals 1 mA and the current gain (βe) is 200 calculate the collector current**<br>$\beta = \frac{I_C}{I_B}$ $200 = \frac{I_C}{0.001}$ $I_C = 0.2\ A$ | |
| | دور ثاني ٢٠٢٤ | |
| 47<br>48 | From the opposite figure: Find the value of $I_C$<br>$I_E = I_B + I_C$<br>$100 = 2 + I_C$<br>$I_C = 100 - 2 = 98\ mA$<br><br> | |

## Compare between each of the following

| No. | Point of comparison | | | | | | |
|---|---|---|---|---|---|---|---|
| | دور اول ٢٠٢٣ | | | | | | |
| 45<br>46 | Point of comparison | NOR gate | | | NAND gate | | |
| | Complete the truth table | Input A | Input B | Output | Input A | Input B | Output |
| | | 1 | 1 | 0 | 1 | 1 | 0 |
| | دور ثاني ٢٠٢٣ | | | | | | |
| 37<br>38 | Point of comparison | P-type semiconductor | | | N type semiconductor | | |
| | type of impurity | Trivalent atoms | | | Pentavalent atoms | | |
| | دور ثاني ٢٠٢٤ | | | | | | |
| 37<br>38 | Point of comparison | NOR gate | | | NAND gate | | |
| | Complete the truth table | Input A | Input B | Output | Input A | Input B | Output |
| | | 1 | 1 | 0 | 1 | 1 | 0 |
| 39<br>40 | Point of comparison | In the forward connection | | | In the reverse connection of P-N junction | | |
| | Electric Resistance of P-N junction | Trivalent atoms | | | Pentavalent atoms | | |

## Correct the underlined in the following statement

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٣ | |
| 28 | <u>Decrease</u> the temperature is one of the methods to increases the electric conductivity of semiconductors | Increase |

| | | |
|---|---|---|
| | دور ثاني ٢٠٢٣ | |
| 27 | In the <u>reverse</u> connection the diode should have a small resistance hence a large current will flow through the diode | Forward |
| 29 | The emitter current is <u>smaller than</u> the sum of the collector current and the base current | Equal |
| | دور ثاني ٢٠٢٤ | |
| 31 | A semiconductor crystal is doped with impurities from pentavalent element so the concentration of free electrons (n) is <u>smaller than</u> the concentration of holes (p). | Bigger than |
| 32 | <u>Digital electronics</u> deal with natural quantities and convert them to continuous electrical signals). | Analog electronics |
| 33 | in transistor the ratio between the collector current and the base current is <u>less than</u> one | More |
| 34 | in thermal dynamic equilibrium the number of bonds broken per second will be <u>greater</u> the number of bonds mended per second. | Equal |

## Give a reason for each of the following

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٤ | |
| 36 | At very low temperature, pure silicon crystal is an insulator | Because all bonds in the crystal are unbroken) and has a high resistance to current flow (no free electrons at 0K) |
| 37 | The thickness of the base of the bipolar junction transistor must be very small | To decrease the chance of recombination OR to decrease the portion lost in the base by recombination |

## A) Put sign (√ ) in front of the correct statement and sign (X) in front of the incorrect statement of each of the following:

| No. | Question | Answer |
|---|---|---|
| | دور اول ٢٠٢٤ | |
| 38 | In an NPN transistor, the emitter is doped with a higher concentration of impurities than the collector | True |
| 39 | The resistance of PN junction obeys Ohm's law | False |
| 40 | In NPN transistor, the collector current is 10mA. If 90% of the electrons emitted reach the collector, then the emitter current will be 11.3 Ma | False |

## Write the name of the logic gate in each of the following cases

| No. | Question | Answer |
|---|---|---|
| | دور ثاني ٢٠٢٤ | |
| 35 | Logic gate has one input. | NOT gate |
| 36 | Logic gate has two inputs and gives high output if the voltage of one of them is high and the other is low | OR gate |